

Unit 5: Human Health, Nutrition, and Disease

Complete Study Notes – Grade 9 Biology

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5.1 What is Food?

Food is any beneficial substance that is eaten, drunk, or otherwise taken into the body to sustain life, provide energy, promote growth, etc. It is consumed to provide nutritional support for an organism. Food is usually of plant, animal, or fungal origin, and contains essential nutrients.

The importance of food in living things:

- **Growth:** Food provides substances necessary for making new cells, tissues, and organs.
- **Energy:** Living things undergo different biological and chemical reactions. These reactions require energy, which comes from food.
- **Replacement of damaged tissues:** Food is required for making new cells to replace damaged ones.
- **Protection from diseases:** Different types of diseases are caused by the shortage of specific nutrients. Food provides protection against deficiency diseases.

5.2 Nutrition

Nutrition is the process of taking in food and converting it into energy and other vital nutrients required for life. Energy obtained from food is important for different activities of the body. The food we eat keeps us alive and provides the nourishment for growth, repair of our body cells, and maintenance of good health.

5.3 Nutrients

Nutrients are important chemical substances that are found in foods. We need nutrients for growth, a source of energy, and to stay healthy. We obtain these nutrients from the foods we eat.

The six important classes of nutrients are: **carbohydrates**, **proteins**, **lipids (fats and oils)**, **vitamins**, **minerals**, and **water**.

Macronutrients vs Micronutrients:

- **Macronutrients** – Required in large quantities: carbohydrates, proteins, lipids.
- **Micronutrients** – Required in small quantities: vitamins and minerals.

Carbohydrates

Carbohydrates provide energy for the human body. They are composed of carbon, hydrogen, and oxygen. One gram of carbohydrate provides, on average, **16 kJ** of energy.

Sources: potatoes, bread, maize, rice, cereals, sugar, fruits, vegetables.

Types: **monosaccharides** (glucose, fructose), **disaccharides** (sucrose, maltose, lactose), and **polysaccharides** (starch, glycogen, cellulose).

Dietary fibre (roughage) is an indigestible carbohydrate found in plant cell walls (cellulose). It adds bulk to the faeces, prevents constipation, and keeps the colon healthy.

Fats and Oils (Lipids)

Lipids are high-energy nutrients providing **37 kJ** per gram. They are composed of carbon, hydrogen, and oxygen. Fats are solid at room temperature (animal sources); oils are liquid (plant sources).

Sources: meat, milk, cheese, butter, egg yolk, sunflower seeds, groundnuts.

Functions: energy store, insulation, protection of organs, absorption of fat-soluble vitamins (A, D, E, K).

Proteins

Proteins provide growth and repair of tissues. They are made of amino acids and contain carbon, hydrogen, oxygen, and nitrogen. One gram provides **17 kJ**.

Sources: meat, fish, eggs, milk, cheese, beans, chickpeas, soybeans, groundnuts.

Functions: growth, tissue repair, enzymes, hormones, antibodies, structural components.

Vitamins

Vitamins are organic substances needed in small quantities for chemical reactions in the body. They do not provide energy but are essential for metabolism.

- **Vitamin A** – Sources: liver, carrots, green vegetables. Function: vision, skin health. Deficiency: night blindness.
- **Vitamin B-complex** – Sources: whole grains, meat, legumes. Function: energy metabolism, nervous system.
- **Vitamin C** – Sources: citrus fruits, tomatoes, green vegetables. Function: tissue repair, immunity. Deficiency: scurvy.
- **Vitamin D** – Sources: fish oil, milk, sunlight (made in skin). Function: calcium absorption, bone health. Deficiency: rickets.

Minerals

Minerals are inorganic substances needed in small amounts for various body functions.

- **Calcium** – Sources: milk, cheese, green vegetables. Function: bones, teeth, muscle contraction, nerve impulses. Deficiency: rickets, osteoporosis.
- **Iron** – Sources: red meat, liver, kidney, eggs, *Tikur teff*. Function: haemoglobin formation (oxygen transport). Deficiency: anaemia.
- **Iodine** – Sources: iodized salt, seafood. Function: thyroid hormone production. Deficiency: goitre.
- **Phosphorus** – Sources: milk, meat, fish. Function: bone formation, energy metabolism.

Water

Water is the major component of body fluids (blood, lymph, tissue fluid). It acts as a solvent and transport medium for substances throughout the body. It also helps regulate body temperature and is essential for all metabolic reactions.

5.4 Balanced Diets

A **balanced diet** is one that contains all of the nutrients in the correct amount and proportion. It must contain enough carbohydrates and fats to meet energy needs, enough protein for growth and repair, and adequate vitamins, minerals, fibre, and water for normal functioning.

Factors affecting dietary needs:

- **Age:** Children and teenagers need more energy and protein for growth. Elderly people need fewer calories but still require a balanced diet.
- **Sex:** Males generally have higher energy requirements due to more muscle mass. Females need more iron during menstruation and pregnancy.
- **Activity level:** Athletes and manual workers need more energy and protein. Office workers have lower energy needs.
- **Physiological state:** Pregnant and breastfeeding women need extra nutrients (calcium, iron, folic acid) for fetal development and milk production.

Example – Energy requirements (kilocalories/day):

Group	Male	Female
8-year-old child	2031	2031
Teenager, aged 14	2987	2318
Adult office worker	2629	2342
Adult manual worker	3585	2987
Pregnant woman	-	2390
Breast-feeding mother	-	2748

5.5 Deficiency Diseases

A **deficiency disease** occurs when a person does not have enough of a particular nutrient and suffers health problems as a result.

Marasmus

Caused by inadequate carbohydrate and energy intake. Common in children. Symptoms: thin arms and legs, little muscle, old-looking face, extremely thin with reduced fat and muscle tissue. Treatment: energy-rich, balanced diet.

Kwashiorkor

Caused by inadequate protein intake. Common in children. Symptoms: dry or flaky skin, swelling of legs and abdomen, changes in hair colour, weakness, irritability. Treatment: protein-rich diet.

Anaemia

Caused by iron deficiency. Results in insufficient haemoglobin, leading to reduced oxygen transport. Symptoms: feeling weak, tired, irritable. Treatment: iron supplements, iron-rich foods (red meat, liver, *Tikur teff*).

Rickets

Caused by vitamin D deficiency. Results in defective bone mineralization, soft bones, bowed legs. Vitamin D is obtained from food (fish, butter, eggs) and from sunlight exposure on skin. Treatment: vitamin D supplements, sunlight.

Scurvy

Caused by vitamin C deficiency. Vitamin C is needed for collagen formation. Symptoms: bleeding under the skin, bleeding gums, poor wound healing. Treatment: vitamin C-rich foods (citrus fruits,

tomatoes, green vegetables).

Summary table of deficiency diseases:

Disease	Deficient Nutrient	Main Symptoms	Sources
Marasmus	Carbohydrate/Energy	Wasting, thin limbs	Energy-rich foods
Kwashiorkor	Protein	Swelling, skin changes	Protein-rich foods
Anaemia	Iron	Tiredness, weakness	Red meat, liver, <i>Tikur teff</i>
Rickets	Vitamin D	Bowed legs, soft bones	Fish oil, sunlight
Scurvy	Vitamin C	Bleeding gums, poor healing	Citrus fruits, vegetables

5.6 Malnutrition

Malnutrition is defined as the insufficient, excessive, or imbalanced consumption of nutrients, which leads to health problems.

The dual burden of malnutrition consists of both **under nutrition** and **overweight/obesity**, as well as diet-related non-communicable diseases.

Under nutrition – most common form, caused by deficiency of protein and dietary energy. Leads to:

- **Wasting** – low weight-for-height; indicates recent and severe weight loss.
- **Stunting** – low height-for-age; chronic undernutrition, irreversible cognitive and physical damage.
- **Underweight** – low weight-for-age.
- **Micronutrient deficiencies** – lack of vitamins and minerals essential for body functions.

Obesity – caused by an imbalance between energy intake and energy expenditure. High intake of fatty foods, added sugars, and too little exercise. Obesity increases the risk of heart disease, high blood pressure, diabetes, and arthritis.

Body Mass Index (BMI):

$BMI = \text{body mass (kg)} / \text{height}^2 (\text{metres}^2)$.

- < 18.5 – underweight
- $18.5\text{--}24.9$ – normal weight
- $25.0\text{--}29.9$ – overweight
- ≥ 30.0 – obese

Example: A person with mass 80 kg and height 1.85 m: $BMI = 80 / (1.85)^2 = 23.4$ (normal weight).

5.7 Substance Abuse

A **drug** is any substance taken into the body that modifies or affects chemical reactions in the body. Substance abuse is using legal or illegal drugs in the wrong way or in excess amounts.

1. Cigarette Smoking

Tobacco smoke contains three main harmful substances: **tar**, **nicotine**, and **carbon monoxide**.

- **Tar** – black, sticky substance that accumulates in the lungs; causes chronic bronchitis, lung cancer, and emphysema.
- **Nicotine** – addictive stimulant that increases heart rate and blood pressure; narrows blood vessels.
- **Carbon monoxide** – poisonous gas that combines with haemoglobin, reducing oxygen transport; damages blood vessels, leading to heart disease.

Chronic bronchitis – tar irritates bronchioles, increases mucus production, cilia damaged, leading to infections and breathing difficulty.

Emphysema – weakened alveoli burst, reducing surface area for gas exchange, causing breathlessness.

2. Drinking Alcohol

Alcohol is a depressant that slows down the nervous system. Short-term effects: loss of coordination, loss of self-control, slower reaction times. Long-term effects: liver damage (cirrhosis), stomach ulcers, brain damage, heart disease. Alcohol is addictive; withdrawal causes tension and irritability.

3. Chewing Khat

Khat (*Catha edulis*) contains **cathinone**, a stimulant that affects the central nervous system. Short-term: feelings of joy, alertness, stimulation. Long-term: depression, irritability, loss of appetite, gastritis, peptic ulcers, difficulty sleeping. Social problems: family breakdown, multiple sexual practices, spread of STIs due to unprotected sex.

4. Cannabis

Cannabis (marijuana) contains THC (tetrahydrocannabinol), a psychoactive compound. Effects: enjoyment, altered time perception, difficulty concentrating, reduced short-term memory, increased appetite. High doses: nervousness, panic, hallucinations, psychosis. Cannabis is illegal in many countries.

5. Heroin

Heroin is a powerful depressant that mimics endorphins in the brain. It causes intense euphoria followed by addiction. Overdose can lead to respiratory failure and death. Sharing needles increases the risk of HIV and hepatitis.

6. Doping in Sports

Doping refers to the use of prohibited substances or methods to improve athletic performance. Common doping agents: anabolic steroids, stimulants, EPO, human growth hormone. Consequences: health risks (heart attack, stroke, liver damage, infertility), unfair advantage, legal penalties, loss of medals, and lifelong bans.

5.8 Infectious and Noninfectious Diseases

A **disease** is a condition in which the body does not function normally, producing symptoms such as headache, fever, pain, or weakness.

5.8.1 Infectious Diseases

Infectious diseases are caused by disease-causing organisms (pathogens) and can be transmitted from person to person.

HIV/AIDS

Caused by the Human Immunodeficiency Virus (HIV). HIV infects and destroys T-helper cells, weakening the immune system. AIDS is the final stage when opportunistic infections occur. Transmission: unprotected sex, blood transfusion, sharing needles, mother-to-child (pregnancy, birth, breastfeeding). Prevention: abstinence, condoms, testing, antiretroviral therapy (ART) for infected mothers. No cure, but ART prolongs life.

Tuberculosis (TB)

Caused by *Mycobacterium tuberculosis*. Bacteria attack the lungs. Transmission: airborne droplets from coughs/sneezes. Symptoms: cough, chest pain, shortness of breath, fever, sweating, weight loss. Prevention: BCG vaccine, good ventilation, avoiding overcrowding, covering mouth when coughing. Treatment: antibiotics (multiple drugs) for 6–9 months.

Malaria

Caused by *Plasmodium* protozoa, transmitted by the bite of female *Anopheles* mosquitoes. The parasite multiplies in red blood cells, causing fever, chills, sweating, anaemia. Prevention: mosquito nets, insect repellents, insecticides, anti-malarial drugs (e.g., artemisinin combination therapy). Chloroquine resistance is widespread.

COVID-19

Caused by the SARS-CoV-2 virus (coronavirus). Spreads via respiratory droplets and aerosols. Symptoms: fever, cough, tiredness, loss of taste/smell, shortness of breath. Prevention: mask-wearing, hand hygiene, physical distancing, vaccination. Pandemic declared in March 2020.

5.8.2 Non-infectious Diseases

Non-infectious diseases are not caused by pathogens. They may be due to genetics, lifestyle, environment, or malnutrition. Examples: deficiency diseases (already covered), heart disease, diabetes, cancer, sickle cell anaemia, allergies, podocytosis. Prevention: healthy diet, exercise, avoiding smoking and alcohol, regular check-ups.

5.9 Renowned Nutritionists in Ethiopia

Nutritionists are researchers or scientists who study nutrition and its impact on health. Ethiopia has several respected professionals in the field of nutrition, contributing to food security, public health, and the fight against malnutrition.

Some notable Ethiopian nutritionists include (as per the textbook, students are encouraged to research and list them). Examples: Dr. **Mesfin Taddese**, Professor **Sebebe Demissew** (also a taxonomist), and others who have worked on nutritional assessment, dietary guidelines, and food fortification in Ethiopia.

Their work includes:

- Identifying nutrient deficiencies and their causes.
- Developing strategies for food fortification (e.g., iodised salt).
- Promoting breastfeeding and maternal nutrition.

- Researching the nutritional value of indigenous Ethiopian foods.
- Working with the government and NGOs to combat malnutrition.

Unit 5 Summary

- **Food** provides energy, growth, repair, and protection from diseases.
- **Nutrients** include carbohydrates (energy), proteins (growth/repair), lipids (energy/storage), vitamins (metabolism), minerals (bone/blood), and water.
- A **balanced diet** contains all nutrients in correct proportions; needs vary with age, sex, activity, and physiological state.
- **Deficiency diseases** arise from lack of specific nutrients: marasmus (energy), kwashiorkor (protein), anaemia (iron), rickets (vitamin D), scurvy (vitamin C).
- **Malnutrition** encompasses both undernutrition (wasting, stunting) and overnutrition (obesity). BMI is used to assess weight status.
- **Substance abuse** includes smoking (tar, nicotine, carbon monoxide), alcohol (depressant), khat (cathinone stimulant), cannabis (THC), heroin, and doping in sports.
- **Infectious diseases** are caused by pathogens: HIV/AIDS (virus), TB (bacteria), malaria (protozoan), COVID-19 (virus). Prevention includes vaccination, hygiene, and vector control.
- **Non-infectious diseases** are not contagious; they include deficiency diseases, heart disease, diabetes, and cancer – often linked to lifestyle.
- **Renowned nutritionists** in Ethiopia have contributed to improving food security and public health.

End of Unit 5 – Human Health, Nutrition, and Disease

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